Enterprise Zero Trust Architecture – 3 Phase Implementation

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Project Overview

This project demonstrates the design and implementation of an enterprise-grade **Zero Trust Security Architecture** using Microsoft Azure and Microsoft 365 security stack. The architecture focuses on identity-first security, device trust enforcement, secure application access, and centralized threat monitoring.

The solution follows a **3-phase implementation approach** aligned with modern Zero Trust principles.

Objectives

- Implement identity-driven security model
- Enforce device compliance and trust
- Enable secure application access without VPN
- Centralize security monitoring and incident response
- Reduce attack surface using Zero Trust principles

Core Technologies

- Microsoft Entra ID (Azure AD)
- Microsoft Intune
- Microsoft Defender for Endpoint
- Microsoft Sentinel (SIEM & SOAR)
- Azure Virtual Network & Security Services
- Microsoft 365 Security Stack

Architecture Approach (3 Phases)

Phase 1: Identity, Device & Data Protection

- Entra ID authentication (MFA / Passwordless)
- Intune device compliance policies
- Windows Autopilot provisioning
- Conditional Access enforcement
- Secure access to SharePoint Online

Phase 2: Secure Application Access

- Azure VM-hosted internal application
- Microsoft Entra Application Proxy
- VPN-less secure access model
- Device compliance + risk-based access control

Phase 3: Security Monitoring & SOC

- Microsoft Sentinel SIEM deployment
- Log ingestion from Entra ID, Defender, Intune, M365
- Automated incident detection and response (SOAR)
- Threat analytics and correlation rules

End-to-End Access Flow (Zero Trust Model)

1. User powers on device
2. Windows Autopilot provisioning starts
3. User authenticates via MFA / Passwordless
4. Device joins Entra ID
5. Device enrolled in Intune
6. Compliance policies applied
7. Conditional Access evaluated
8. Application access granted via App Proxy
9. Logs sent to Microsoft Sentinel
10. Automated detection & response triggered

Security Controls Implemented

- Multi-Factor Authentication (MFA) enforced
- Passwordless authentication (FIDO2 / Windows Hello)
- Device compliance required for access
- Legacy authentication blocked
- BitLocker & Microsoft Defender enabled
- Conditional Access with risk-based policies
- No public exposure of internal applications

Security Outcomes

✓ Zero Trust architecture implemented end-to-end
✓ Secure VPN-less application access achieved
✓ Identity-driven access control enforced
✓ Real-time threat detection via Sentinel
✓ Automated incident response (SOAR workflows)
✓ Centralized cloud-native security monitoring (SOC-ready environment)

Key Security Principles Applied

- **Verify Explicitly** → Continuous authentication & evaluation
- **Least Privilege** → Minimal access based on role & device compliance
- **Assume Breach** → Continuous monitoring & incident response

Conclusion

This architecture demonstrates a scalable and enterprise-ready Zero Trust implementation using Microsoft security ecosystem. It provides strong identity security, secure device management, application protection, and centralized SOC capabilities aligned with modern cloud security standards.